

[Pipeline Suite] AOV Rebuild Pipeline

Master Suite - User Manual

This tool is designed for advanced digital compositing pipelines. It provides a one-click automated workflow to intelligently unpack, analyze, and reconstruct the core multi-channel EXR Beauty images from major render engines (Arnold, Redshift, V-Ray, Karma).

1. Prerequisites

1. **Node Selection Required:** You **must select a valid Read node** in Nuke's Node Graph before executing the script.
2. **Multi-channel EXR Support:** The chosen Read node must contain rich AOV (layer) metadata rendered out from 3D software. If no node or an invalid node type is selected, an error prompt will block execution.

2. Core Operational Workflow

Step 1: Trigger Tool & Language Choice

- Select your Read node and press the hotkey **Alt + Shift + R** (or go to the top menu bar: Pipeline -> AOV Rebuild Master Suite).
- **First Prompt:** A language selector window will pop up. Choose **English** or **chinese**.

Step 2: Parameter Panel Configuration

After choosing the language, the Master Suite UI panel will pop up. Configure the options based on your asset profile:

- **Rebuild Mode:**
 - **Basic Standard:** Extract essential AOV contribution layers (such as Diffuse, Specular, etc.) and merge them via **Plus (Add)**. Ideal for standard prompt shot look-dev adjustments.
 - **Advanced:** Activates advanced photon-splitting workflows. For Redshift and V-Ray, it isolates Raw lighting passes and Filter/Tint passes, re-combines them using **Multiply**, and then feeds them into the main global **Plus** trunk.

- **Renderer Engine:**
 - **Auto Detect (Highly Recommended):** The tool automatically reads internal EXR metadata fields to instantly identify if the source file was generated via Arnold, Redshift, V-Ray, or Houdini Karma.
 - **Manual Override:** If the metadata header has been stripped, you can manually force lock the system to a specific renderer blueprint.
- **Run Quality Control (QC_mode):**
 - Check this box to automatically deploy a delta check comparison framework at the bottom of the node hierarchy, producing Report.

3. Node Graph Procedural Rules

Upon execution, the script mathematically lays out an organized matrix of nodes underneath your source image:

- **Dot Node Bus:** Creates horizontal signal paths on top to preserve a clean pipeline appearance.
- **Shuffle2 Unpacking:** Every isolated AOV is split using Nuke's robust Shuffle2 node architecture, with Postage Stamp previews enabled for high glanceability.
- **Merge2 Calculations:**
 - Basic elements and Direct/Indirect passes for Karma/Arnold are stacked sequentially using the **plus** operation.
 - Advanced Raw and Filter pairs are locked via **multiply** math before merging into the main pipeline stream.

4. Interpreting the QC Audit Report (With QC Mode Enabled)

Once you click OK with QC mode activated, Nuke will attach your active Viewer directly to the newly generated **QC_Difference_Check** node (set to difference mode) and display a detailed diagnostic message:

1. **EXR Layer Ingredient Verification:**
 - **Green [EXR Layers Complete]:** The source file includes all necessary ingredients required by the current rendering blueprint.

Reconstructed images will theoretically achieve a **100% mathematical match** to the original Beauty pass.

- **Orange [EXR Layer Warning]:** Explicitly lists missing essential components. This indicates that the Difference node may yield residual artifact errors due to an incomplete recipe.

2. Extraction Node Integrity Check:

- **Green [Node Extraction OK]:** All identified layers successfully compiled corresponding Shuffle nodes.
- **Red [Node Extraction Error]:** Highlights any expected channels that failed to generate nodes, simplifying studio workflow troubleshooting.